

T4EDR: HYBRID THREAT DETECTION FRAMEWORK FOR EDR BASED ON SEMANTIC RULE EMBEDDING AND CONTEXTUAL NETWORK FLOW ANALYSIS

Muh Dzikri Alfauzan Nuzul¹, Desi Anggreani¹, Muhammad Faisal¹,
Abd Rahman Wahid^{1,*}, Titik Khawa Abd Rahman²

¹Departement of Informatics, Universitas Muhammadiyah Makassar, Indonesia

²Department of School of Science and Technology, Asia E University, Malaysia

Abstract

In response to the increasing complexity of cyber threats and attacks, this study proposes an innovative endpoint detection and response (EDR) framework named T4EDR (Transformer for Endpoint Detection and Response). Specifically, it addresses the inefficiencies of traditional systems that rely on static and less adaptive rules. T4EDR integrates semantic analysis methods to interpret security rules and a transformer-based deep learning model to contextually analyze network traffic flows. In the initial phase, thousands of security rules from Wazuh were extracted, embedded, and semantically validated against the MITRE ATT&CK framework, achieving a semantic coherence of 86.43% with a silhouette score of 0.702. Subsequently, the FlowBERT model was designed to classify network traffic flows using the CIC-IDS2018 dataset, achieving 91.1% accuracy, a macro-F1 of 0.79, and a mean Average Precision (mAP) of 0.90, surpassing the quantitative target of 85%. The integration of rule embeddings with FlowBERT hidden states through a linear projector enables adaptive mapping of endpoint activities to relevant security rules, supporting context-based automated responses. The main contribution of this study is an adaptive framework that bridges the gap between traditional rule analysis and deep learning-based detection, thereby enhancing the capability to detect multi-stage threats on modern endpoints.

Keywords: Cyber Security, Deep Learning, Endpoint detection and Response, Transformer, Rule Embedding

Received: 29-09-2025 | Accepted: 27-02-2026 | Available Online: 31-03-2026

DOI: <https://doi.org/10.23887/janapati.v15i1.104059>

I. INTRODUCTION

The ability to proactively detect and respond to cyber threats is a key pillar in maintaining the integrity and security of modern technology ecosystems. The increasing complexity of threats and multi-stage attacks specifically targeting endpoints as strategic entry points highlights this urgency [1]. Recent industry reports indicate that more than 70% of cyberattacks originate from endpoint compromise, which then escalate into lateral exploitation and data exfiltration [2]. Under such conditions, signature-based detection proves insufficiently adaptive as it heavily relies on manually updated threat databases [3].

Academic research in the fields of network-based intrusion detection systems (NIDS) and host-based intrusion detection systems (HIDS) has leveraged machine learning and deep learning to improve attack classification accuracy

[4, 5]. Deep learning methods such as Convolutional Neural Networks (CNN) [6] and Long Short-Term Memory (LSTM) [7] have demonstrated strong performance in detecting temporal anomalies [8]. Nevertheless, further studies reveal that these models struggle when faced with new data distributions (domain shift) and often behave as black-box systems, making it difficult to align their outputs with standard security frameworks [9, 10].

On the other hand, rule-based approaches employed in platforms such as Wazuh provide semantic clarity due to their reliance on explicit rules [11]. However, such systems face limitations in detecting novel attacks, lack scalability, and are prone to generating false positives [12]. This underscores a gap between rule-based methods, which are interpretable but static, and deep learning approaches, which are adaptive but opaque [13].

*Corresponding author: 105841116522@student.unismuh.ac.id (A. R. Wahid)

As research advances, Transformers have emerged as a promising approach for cyber threat detection. FlowTransformer models demonstrate the ability to capture long-term dependencies in network traffic [14], while Transformer-based self-supervised contrastive learning improves generalization across heterogeneous datasets [15]. Other studies combine CNN, BiLSTM, and Transformer architectures for IoT intrusion detection, achieving up to 99% accuracy [16]. Similar trends are evident in research leveraging Transformer-based IDS for malware and advanced persistent threats [17, 18].

The integration of Endpoint Detection and Response (EDR) with standardized frameworks such as MITRE ATT&CK is also gaining attention, as it enables mapping of detection outcomes to attacker tactics, techniques, and procedures (TTPs) [19, 20]. Moreover, the demand for Explainable AI (XAI) in cybersecurity continues to grow, ensuring that detection systems are both accurate and interpretable for security analysts [21, 22]. Recent studies also highlight the importance of employing modern datasets such as CIC-IDS2018 in evaluating deep learning-based IDS [23], as well as adapting IDS to IoT and cloud environments [24, 25].

The novelty of this study lies in explicitly integrating security rule representations with Transformer-based representation learning, which can improve multi-stage attack detection accuracy to above 85% and ensure rule semantic coherence exceeding 80%, an aspect rarely explored in current literature.

II. METHOD

A. Overall Framework Architecture

The T4EDR architecture is designed as a modular and interpretable framework that integrates explicit rule-based knowledge with the adaptive representation power of Transformer models. Conceptually, this framework bridges the limitations of static rule-based detection which often lacks flexibility and contextual awareness with the deep contextual modeling capabilities of Transformer-based systems. The architecture reflects the modern evolution of Endpoint Detection and Response (EDR) systems, where explainability, adaptability, and semantic alignment are equally critical for cyber defense effectiveness. At the input layer, security rules from OSSEC/Wazuh are extracted and transformed into structured representations containing match patterns, severity levels, compliance metadata, and associations with the

MITRE ATT&CK taxonomy [26]. These attributes encapsulate expert-defined operational semantics and form the primary knowledge base of the system. The structured rule data then undergo normalization, categorical encoding, and TF-IDF based text representation to ensure numerical stability and textual feature richness suitable for machine representation learning. The architecture is organized into five hierarchically connected phases data processing, rule embedding, contextual analysis, adaptive enforcement, and adaptive response as illustrated in Figure 1. Each phase contributes a distinct functional layer while maintaining logical continuity within the detection response pipeline.

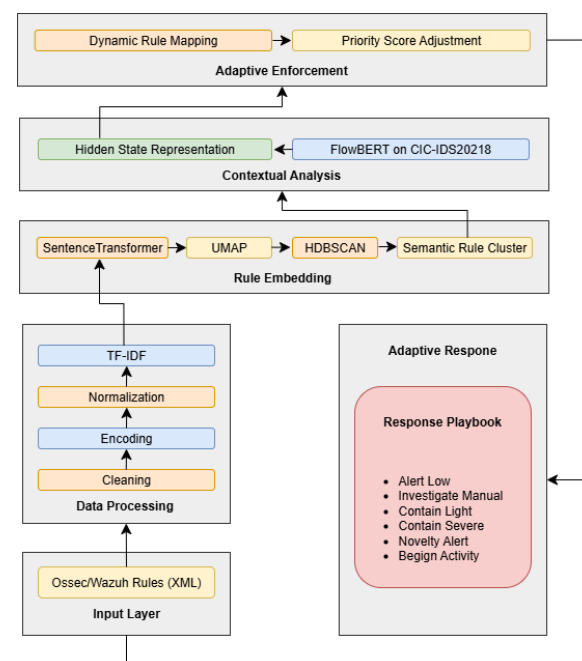


Figure 1. Architecture Flow Diagram of the T4EDR Framework

The rule embedding phase transforms symbolic rules into continuous semantic vectors using a pre-trained SentenceTransformer model. This process embeds linguistic and structural information into a multidimensional space, allowing machine-level reasoning over previously human-readable rules. Subsequent dimensionality reduction with UMAP and clustering with HDBSCAN are applied to preserve local topology and semantic proximity, enabling the formation of clusters that correspond to coherent operational or tactical domains. The contextual analysis phase processes real-world traffic from the CIC-IDS2018 dataset [18], which represents diverse attack scenarios such as DoS, brute force, and infiltration. Using a Transformer variant named FlowBERT, the model learns dynamic

representations of network flows that capture inter-feature dependencies and temporal contextuality [27]. The hidden state representations produced by FlowBERT serve not only as inputs for classification but also as a bridge to align contextual flow information with semantic rule embeddings. In the adaptive enforcement phase, FlowBERT outputs are dynamically mapped to the nearest security rules in the embedding space through nearest-neighbor similarity search. This alignment allows the system to update rule priority scores continuously, rewarding accurate detections, penalizing false positives, and boosting under-represented attack detections. Such adaptivity ensures that the system remains sensitive to evolving threat patterns without relying solely on manual rule updates. Finally, the adaptive response module transforms detection outcomes into actionable security decisions guided by a predefined response playbook. Each decision considers the model's confidence, rule relevance, and uncertainty metrics to determine appropriate actions ranging from low-level alerts to severe containment procedures. Analyst feedback and post-event verification are integrated into a feedback loop, allowing the system to refine thresholds, retrain the model, and recalibrate rule weights over time. This self-improving design embodies a learning-oriented EDR philosophy, aligning with modern perspectives on autonomous yet explainable cybersecurity frameworks. Overall, T4EDR establishes a unified architecture that harmonizes symbolic reasoning with contextual learning. It not only enhances the adaptability of EDR systems but also reinforces semantic transparency and operational auditability key requirements for resilient cyber defense in the era of intelligent automation.

B. Data Processing

The initial phase of this study focused on the collection and mining of security rules from the Wazuh rule-local repository, which consists of hundreds of thematic XML files covering core OSSEC/Wazuh rules, system services, web applications, databases, firewalls, and network devices. To formalize the workflow, the set of source files is denoted as f . A file processing operator $P(\cdot)$ maps an XML file into a set of structured rules, and all outputs are merged into a unified dataset D :

$$D = \bigcup_{f \in \mathcal{F}} p(f) \quad (1)$$

Each rule $r \in D$ is expressed as a labeled tuple:

$$r = (\text{File_Name}, \text{Rule_ID}, \text{Level Match}, \text{Description}, \text{Groups}, \text{MITRE_ID}, \text{If_Sid}, \text{If_Matched_Sid}, \text{Frequency}, \text{Timeframe}, \text{ignore}) \quad (2)$$

Among these attributes, MITRE_ID represents a set of techniques $\{T_1, T_2, \dots\}$, which are serialized for storage. Missing value normalization is applied in the final stage to ensure schema consistency and minimize obstacles during subsequent analysis. The resulting dataset contains 3,056 rule records. The schema is represented in Table 1, which corresponds to the structure of combined_rules_dataset.csv used in this study.

Table 1. Combined rule set scheme

Column	Description
File_Name	Source XML file name of the rule
Rule_ID	Unique identifier of the rule in the source file
Level	Severity or notification level of the rule
Match	Pola pencocokan Match pattern (regex/substring) triggering the rule
Description	Narrative description of the rule and event context
Groups	Class/category of the rule
MITRE_ID	List of referenced MITRE ATT&CK techniques
If_Sid	Dependency on another rule based on SID
If_Matched_Sid	Dependency activated if another rule has been triggered
Frequency	Frequency threshold as a condition/attribute of rule evaluation
Timeframe	Time window for frequency evaluation
Ignore	Exception/ignore marker for specific conditions

Table 2 provides a sample of the dataset used in the study.

Table 2. Combined rule set scheme

File Name	Rule ID	Level	Match	Description	Groups	MITRE ID	If SID
0015-ossec_rules.xml	501	3	Agent started	New wazuh agent connected	pci_dss_10.6.1,gpg13_10.1,gdpr_IV_35.7.d,hipaa_164.312.b,nist_800_53_AU.6,tsc_CC7.2,tsc_CC7.3		500
0015-ossec_rules.xml	502	3	Manager Started	Wazuh server started	pci_dss_10.6.1,gpg13_10.1,gdpr_IV_35.7.d,hipaa_164.312.b,nist_800_53_AU.6,tsc_CC7.2,tsc_CC7.3		500
0015-ossec_rules.xml	503	3	Agent started	Wazuh agent started	pci_dss_10.6.1,pci_dss_10.2.6,gpg13_10.1,gdpr_IV_35.7.d,hipaa_164.312.b,nist_800_53_AU.6,nist_800_53_AU.14,nist_800_53_AU.5,tsc_CC7.2,tsc_CC7.3,tsc_CC6.8		500
0015-ossec_rules.xml	504	3	Agent disconnected	Wazuh agent disconnected	pci_dss_10.6.1,pci_dss_10.2.6,gpg13_10.1,gdpr_IV_35.7.d,hipaa_164.312.b,nist_800_53_AU.6,nist_800_53_AU.14,nist_800_53_AU.5,tsc_CC7.2,tsc_CC7.3,tsc_CC6.8	T1562.001	500

Embedding of detection rules forms the foundation for understanding and grouping thousands of heterogeneous rules into consistent semantic representations. In this study, each rule was mapped into a high-dimensional vector through the SentenceTransformer model (all-MiniLM-L6-v2), which simultaneously processes textual attributes such as match patterns and descriptions, along with numerical features and MITRE ATT&CK-based binarized features. This strategy enables the integration of linguistic and contextual aspects with operational metrics, resulting in richer and security-oriented vector representations. Mathematically, the embedding process is formalized as follows.

C. Rule Embedding

Embedding of detection rules forms the foundation for understanding and grouping thousands of heterogeneous rules into consistent semantic representations. In this study, each rule was mapped into a high-dimensional vector through the SentenceTransformer model (all-MiniLM-L6-v2), which simultaneously processes textual attributes such as match patterns and descriptions, along with numerical features and MITRE ATT&CK-based binarized features. This strategy enables the integration of linguistic and contextual aspects with operational metrics, resulting in richer and security-oriented vector

representations. Mathematically, the embedding process is formalized as follows:

$$E_r = f_{\theta}(M_r, D_r, N_r, A_r) \quad (3)$$

Where :

E_r : Rule embeddings,
 M_r dan D_r : Representations of match pattern and description,
 N_r : Vector of numerical features,
 A_r : Vector of attack binaries,
 f_{θ} : Model representation function trained with parameter θ).

To reduce complexity and enhance pattern interpretability, the 925-dimensional embeddings were projected into a two-dimensional space using UMAP (Uniform Manifold Approximation and Projection). The reduced representations were then clustered using HDBSCAN, which is capable of identifying noise points.

The experimental results demonstrated that from 3,045 rules, a total of 131 unique clusters were formed, with approximately 8.9% of the rules identified as noise. The quality metric, calculated with a Silhouette Score of 0.702, indicated good separability between clusters. Further semantic validation using MITRE ID dominance showed that 121 out of 140 non-noise clusters (86.43%) achieved high coherence, meaning that more than half of the entries within

a cluster shared the same dominant MITRE ID. The embedding process therefore yielded linguistically consistent results and semantically captured tactical similarities referenced in the ATT&CK framework. For example, a fully coherent cluster with MITRE T1562.001 predominantly contained rules related to Defensive Evasion via Disable Security Tools, while the cluster associated with MITRE T1105 represented Ingress Tool Transfer attack patterns. A smaller proportion of clusters (13.57%) were identified as incoherent, reflecting diverse descriptions or contexts of rules that were not fully modeled.

Table 3. Evaluation Summary of Rule Embedding

Metrics	Value	Interpretation
Number of rules	3056	Main dataset
Initial embedding dimension	925	Concatenated multi-feature result
Number of clusters (non-noise)	140	Variation in rule patterns
Noise proportion	8.90%	Outliers or rare rules
Silhouette Score	0,702	High separation quality
Coherent clusters ($\geq 50\%$ MITRE ID)	121 (86,43%)	Semantik konsisten
Incoherent clusters	19 (13,57%)	High semantic variation

The rule embedding stage successfully established a semantic space to support subsequent phases, including rule alignment, traffic correlation, and integration into the FlowBERT/T4EDR model. The embeddings serve not only as vector representations but also as conceptual bridges between traditional rule-based detection and representation-based machine learning approaches.

D. Contextual Analysis

Contextual analysis was performed on network traffic to move beyond static surface patterns toward a more dynamic understanding of attack behaviors. In this study, contextual analysis was applied using the CIC-IDS2018 dataset, which includes diverse attack types such as Botnet, Brute Force, Web/XSS, DoS GoldenEye, Infiltration, and SQL Injection. Each flow record was processed through cleaning, standardization, and removal of irrelevant attributes, leaving only numerical features that could be directly ingested by the Transformer-

based model. Figure 2 illustrates the design architecture of the contextual analysis.

The FlowBERT architecture was adopted and designed to capture the contextual representation of network traffic flows. Preprocessing steps included numerical feature normalization using StandardScaler to balance feature distributions, label encoding into numerical form via LabelEncoder, and class imbalance handling through weighted loss functions.

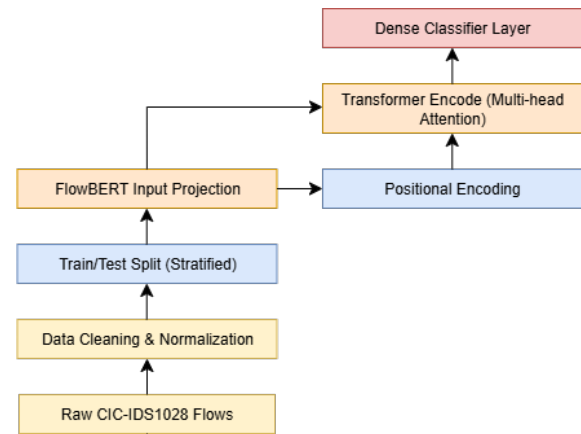


Figure 2. Contextual Analysis Design Architecture

Mathematically, the normalization process is expressed in Formula 4:

$$x'_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j} \quad (4)$$

Where :

x_{ij} : Value of the j -th feature for i -th sample,

μ_r : Mean of the j -th feature,

σ_j : Standard deviation of the j -th feature

The architecture projects feature vectors of dimension d into a latent space of dimension h through a linear layer, adds positional encoding, and processes them with multi-head self-attention within Transformer Encoder layers. The final representations are then classified into one of seven attack classes.

This strategy enables the model to recognize individual characteristics of packets or flows and to capture contextual interactions across features, which often reflect specific attack tactics. With a total of 1.71 million flow records processed into 70 numerical features, the approach ensures consistent large-scale context-aware analysis. The phase concludes with stratified sampling into training and test sets (80:20), preserving class distribution and ensuring that contextual analysis yields stable and representative embeddings.

E. Adaptive Enforcement

This stage is designed to bridge the outcomes of contextual classification with actionable security rules on the EDR platform. The approach leverages the latent representations generated by FlowBERT and dynamically maps them to Wazuh rules through nearest neighbor search in the embedding space. This mechanism enables each classified traffic flow to be linked with the most relevant rule within its semantic cluster. Since the dimensionality of FlowBERT embeddings ($h=128$) is not aligned with the rule embeddings obtained from the rule embedding phase ($d=925$), Principal Component Analysis (PCA) projection is applied. The transformation is expressed as:

$$z_i = P \cdot e_i \quad (5)$$

Where :

e_i : Original rule embedding

P : PCA projection matrix

z_i : Rule embedding projected into FlowBERT space

Once the vector spaces are aligned, each flow representation f_j is mapped to its nearest rule using cosine similarity. This mechanism results in dynamic rule mapping that updates rule priority scores based on prediction quality. The adaptive score S_r for a rule r is updated as:

$$S_r^{(t+1)} = \text{clip}(S_r^{(t)} + \alpha \cdot \delta_{FP} - \beta \cdot \delta_{FN} + \gamma \cdot \delta_{FN}, 0, 1) \quad (6)$$

Additionally, an uncertainty threshold ($\tau = 0.7$) is applied to flag low-confidence predictions, ensuring that false negatives with high uncertainty trigger novelty alerts. This approach strengthens rules proven to be relevant while creating room for detecting attacks not explicitly defined in the existing rule base. Table 3 summarizes the key adaptation parameters used in the simulation.

Table 3. Dynamic Adaptation Parameters

Parameter	Value	Function
α (True Positive Gain)	0.05	Increases rule priority when detection is correct
β (False Positive Penalty)	-0.10	Decreases rule priority when normal traffic is misclassified
γ (False Negative Boost)	0.10	Reinforces rules when attacks are missed as benign
τ (Uncertainty Threshold)	0.10	Triggers novelty alerts for low-confidence predictions

This stage demonstrates the ability of the framework to adjust rule priorities dynamically based on empirical evidence from classification results, reinforcing adaptive enforcement as a core component of T4EDR.

F. Adaptive Response

The adaptive response stage is designed as a decision-making module that operates on the dynamically adjusted rule priority scores from the previous phase. This approach emphasizes risk-aware decision making, where FlowBERT predictions are not directly translated into automated actions but contextualized with model confidence, dynamically mapped Wazuh rules, and adaptive parameters derived from earlier evaluations. Formally, the response decision is defined by the following function:

$$R(e) = f(C_p(e), P_r(r), U(e), A_{log}(e)) \quad (7)$$

Where:

$C_p(e)$: FlowBERT prediction probability for event e ,

$P_r(r)$ Priority score of the relevant rule r produced by adaptive enforcement,

$U(e)$: Prediction uncertainty score ($1 - \text{maximum confidence}$),

δ : Indikator kondisi dalam prediksi

$A_{log}(e)$: Log of prior adaptive actions.

Based on this function, the system refers to a response playbook encoded with predefined actions, represented in Table 4. The playbook spans a spectrum of responses ranging from informational logging, manual escalation to analysts, to automated containment actions such as temporary blocking or host isolation.

Table 4. Parameter Adaptasi Dinamis

Action Code	Dampak	Category
Alert Low	Minimal	Automatic/Manual
Investigate Manual	Low-Medium	Manual
Contain Light	Medium	Semi-automatic
Contain Severe	High	Automatic with conditions
Novelty Alert	High	Manual
Benign Activity	None	Automatic

Response thresholds are layered, with $T_{low} = 0.35$, $T_{med} = 0.65$, $T_{high} = 0.85$, and an additional $C_{auto} = 0.90$ as the minimum requirement for high-risk automated execution. This design ensures the decision engine avoids overreaction to low-confidence predictions while allocating SOC resources proportionally to threat severity.

The response mechanism also incorporates the concept of a feedback loop, whereby execution outcomes, such as containment effectiveness or analyst validation are reintegrated into the system to recalibrate thresholds, adjust rule weights, and enrich the FlowBERT training dataset. With this design, the system gradually strengthens resilience and reduces dependency on full manual intervention.

III. RESULT

A. Rule Corpus and Data Quality

The constructed rule corpus produced 3,056 valid entries consolidated from multiple Wazuh and OSSEC rule families. In terms of quality, the extracted entries preserve semantic connectivity through the Groups, Description, and MITRE ATT&CK taxonomy fields. Numerical normalization of the Level, Frequency, and Timeframe columns yielded a balanced distribution, making the corpus suitable for downstream machine learning tasks. The categorical attribute Groups was consistently encoded, while narrative descriptions were transformed into semantic representations using TF-IDF with 1,000 feature dimensions, enriching the corpus with a linguistic foundation. Figure 3 illustrates the distribution of rule severity levels and the most frequent threat groups.

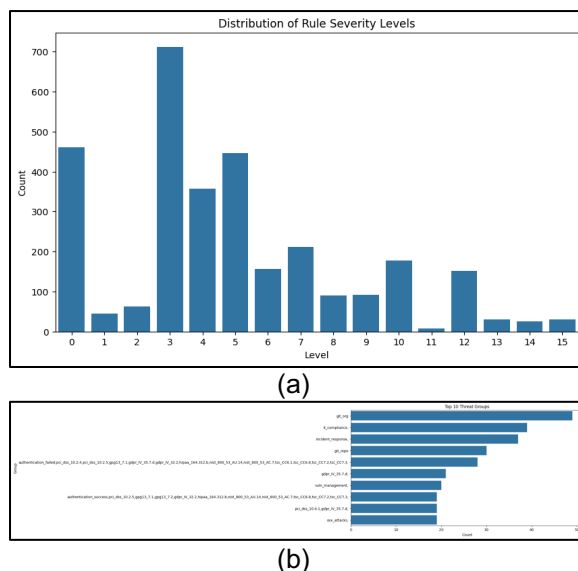


Figure 3. (a) Distribution of Rule Severity Levels, (b) Top 10 Threat Groups

Table 5 summarizes the structure and output of the constructed rule corpus.

Table 5. Summary of the Corpus of Construction Rules

Key Attributes	Amount/Description
Total rule entries	3.056
Number of source files	>120 XML files (multi-domain)
Core columns	12 (File Name, Rule ID, Level, Match, Description, Groups, MITRE ID, If Sid, If Matched Sid, Frequency, Timeframe, Ignore)
Additional features	1,000 TF-IDF features + multi-label MITRE
Data completeness	>95% fields populated after normalization ±20 files (junk after document element, invalid token)
Parsing failures	

B. Semantic Space and Cluster Coherence

The mapping of rules into a representational space yielded a 925-dimensional semantic space which, after projection with 2D UMAP, exhibited a well-segmented structure with only a small fraction of outliers. Across the full set of 3,056 rules, the HDBSCAN algorithm identified 131 cluster labels (including -1 for noise) with 272 rules (8.90%) classified as noise. The quality of separation between clusters was reflected in a Silhouette Score of 0.702, indicating compact intracluster cohesion and adequate intercluster separation. Semantic coherence was assessed using the dominance of MITRE ATT&CK IDs within each non-noise cluster. Out of 140 non-noise clusters, 121 clusters (86.43%) met the coherence criterion ($\geq 50\%$ of entries sharing the same dominant MITRE ID). For illustration, the cluster dominated by T1562.001 predominantly contained defense evasion rules related to disabling security tools, whereas the cluster associated with T1105 represented ingress tool transfer patterns. In contrast, the remaining 19 clusters (13.57%) were incoherent, typically reflecting mixed contexts or diverse rule descriptions that hindered the formation of a single semantic dominance. Compact clusters illustrate homogeneous semantic domains (GitHub activity, Microsoft Graph events, PowerShell processes), while elongated bands reveal content variations that nonetheless preserve semantic continuity.

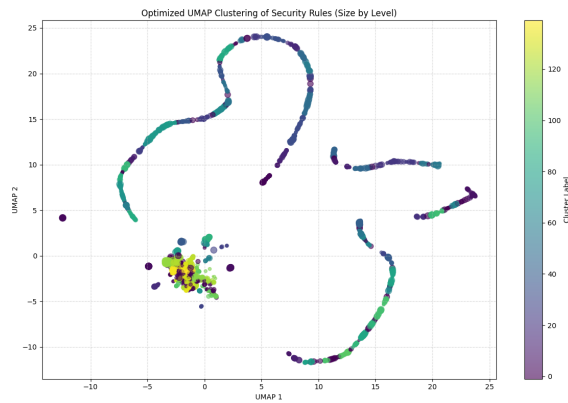


Figure 4. Optimized UMAP Clustering of Security Rules (Size by Level)

C. Contextual Flow Analysis and Classification Performance

The evaluation of the contextual module using FlowBERT on the CIC-IDS2018 test set demonstrated stable training behaviour and strong generalization. The training curves showed a consistent reduction in loss from 0.509 to 0.269 across 20 epochs, while training accuracy remained relatively steady at approximately 0.70, indicating controlled optimization without sharp signs of overfitting. On the held-out test set comprising 342,029 samples, the model achieved accuracy of 0.911, weighted precision of 0.901, weighted recall of 0.911, and weighted F1-score of 0.904. Error mapping through the confusion matrix confirmed highly robust class separation for Bot and DoS GoldenEye (precision, recall, and F1 near-perfect), as well as solid performance for Benign traffic ($P=0.92$; $R=0.96$; $F1=0.94$).

Conversely, Infiltration exhibited a clear trade-off ($P=0.55$; $R=0.38$; $F1=0.45$), with bidirectional misclassification against Benign traffic, reflecting statistical proximity of sessions and the “grey-zone” nature of such flows. Rare classes such as Brute Force –Web ($P=0.29$; $R=0.72$; $F1=0.41$), Brute Force –XSS ($P=1.00$; $R=0.43$; $F1=0.61$), and SQL Injection ($P=0.80$; $R=0.47$; $F1=0.59$) showed high precision but limited recall, indicating that contextual signals for low-frequency events were captured narrowly yet remain partial in coverage. Figures 5, 6, and 7 are graphs of the results obtained from Contextual Flow.

C. Semantic Space and Cluster Coherence

The adaptive enforcement mechanism recalibrates rule priorities contextually based on FlowBERT outputs and semantic proximity to the most relevant Wazuh rules. After simulations on

20,000 test events, the distribution of priority scores shifted markedly from an initial curve concentrated in the low–medium range toward an adapted curve weighted more heavily in the high range, indicating a re-focus of protection toward areas with the greatest impact on detection operations.

The effect was measurable in the average priority scores of rule groups associated with major attack types: Brute Force–Web and Brute Force–XSS increased by approximately 104.55%, DoS GoldenEye by 216.96%, Infiltration by 108.72%, and SQL Injection by 72.85%. The target threshold of $>20\%$ improvement was achieved across all critical classes, demonstrating that score adjustments operated consistently across tactical domains, including low-frequency scenarios that had previously been underrepresented.

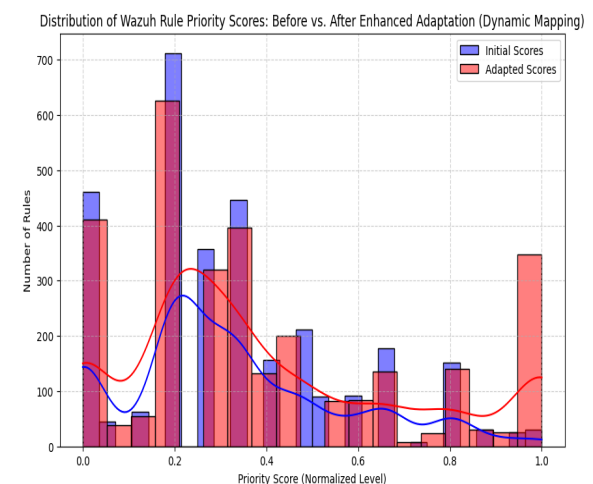


Figure 8. Distribution of Wazuh Rule Priority Score (Before vs After Enhanced Adaption)

In parallel, the module generated 1,170 novelty alerts (predominantly from statistically ambiguous Infiltration flows) and recorded 1,366 adaptive actions, both serving as feedback signals for refining rules and calibrating the model in subsequent iterations. Taken together, the post-adaptation pattern reflects a system that is more risk-seeking toward high-value threats while retaining measured penalties on high-confidence false positives, establishing a healthy balance between sensitivity and operational actionability.

D. Adaptive Response Engine Performance

The execution of the final phase positions the T4EDR framework within the response domain, where model predictions and dynamic rule mappings are translated into operational decisions. Simulations involving 20,000 test events employed a playbook-driven decision

pipeline encompassing a spectrum of actions, ranging from alert low to contain severe.

The severity distribution revealed a dominance of None actions (benign activity), accounting for more than 13,700 cases, consistent with the prevalence of normal traffic in the CIC-IDS2018 dataset. Critical actions, such as contain severe and novelty alert, totaled 5,568 cases, reflecting the system's ability to escalate significant threats. Meanwhile, medium-level responses (manual investigation and light containment) were relatively minor, suggesting that most decisions fell into two extremes, benign activity or full containment, as depicted in Figure 9.

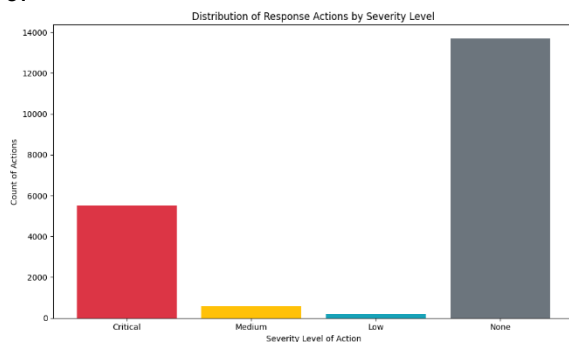


Figure 9. Distribution of Response Actions by Severity Level

Further mapping of individual threat classes highlighted patterns aligned with their detection complexity. The Bot class dominated contain severe actions, mirroring its deterministic nature and high F1 score. In contrast, Figure 10 shows the Infiltration class producing a more diverse spectrum of responses, ranging from alert low to manual investigation and novelty alerts. This behavior aligns with its high uncertainty and the potential for zero-day like exploitation within infiltration scenarios.

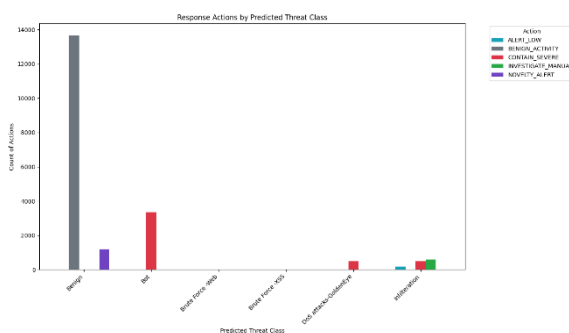


Figure 10. Response Actions by Predicted Threat Class

The performance evaluation indicated a simulated mitigation success rate of 75.13% (Figure 11), meaning that three out of four attack incidents were effectively neutralized through adaptive response execution. Conversely, the false-positive response rate of 8.78% remained within an acceptable range for an experimental

system, while signalling the need for feedback-driven refinement and further tuning. These results confirm that the adaptive response module successfully translates FlowBERT analysis and rule embeddings into practical actions, striking a balance between automation and manual intervention. This equilibrium is crucial to maintaining rapid reaction times while reducing the operational risks of misclassification.

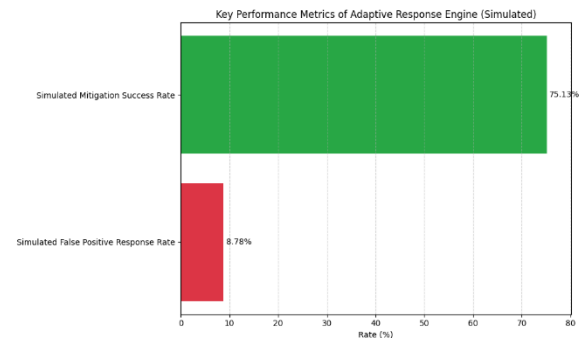


Figure 11. Key Performance Metrics of Adaptive Response Engine (Simulated)

The overall outcomes demonstrate how the integration of rule embedding, contextual flow analysis, dynamic mapping, and adaptive response establishes a more contextual, transparent, and resilient EDR framework. Building on this foundation, the subsequent discussion will synthesize the key achievements, acknowledge system limitations, and outline directions for future development.

IV. DISCUSSION

The findings indicate that aligning the semantic representation of security rules with the contextual representation of network flows significantly enhances the actionability of EDR systems. The coherence of MITRE-based clusters ($\approx 86\%$) ensures that rule embeddings retain their taxonomic meaning, while the accuracy of FlowBERT on the CIC-IDS2018 dataset (≈ 0.91) underscores its generalization capacity for multi-stage attack patterns. The combination of these components through dynamic mapping and adaptive score adjustment yields an adaptive response distribution: the majority of normal traffic is automatically classified as benign activity, whereas high-value attacks trigger containment or novelty alerts. This pattern not only reduces the workload for analysts on routine events but also accelerates escalation when critical risks emerge. Operationally, a simulated mitigation success rate above 75% and a false positive rate below 10% demonstrate that the T4EDR framework

meets practical thresholds for a reliable detection response system.

Nevertheless, several limitations remain. The PCA-based projection used to align rule embeddings with FlowBERT's latent space is inherently linear, potentially overlooking non-linear feature relationships, while approximately 14% of incoherent clusters reflect annotation ambiguities within the rules. Evaluation has also been limited to public benchmark datasets, necessitating field validation on heterogeneous production logs. To address these issues, future enhancements should integrate non-linear learned projectors, incorporate active learning mechanisms driven by novelty alerts, and apply reinforcement learning within the response module to ensure continuous adaptation.

In this light, the main contribution of T4EDR lies in unifying the semantic transparency of rule-based systems with the adaptivity of contextual learning. This synthesis theoretically shortens time-to-respond while simultaneously strengthening the foundation of auditable security governance.

V. CONCLUSION

This research has successfully developed T4EDR, a Transformer-based EDR framework that integrates rule embedding, contextual flow analysis, and adaptive enforcement and response mechanisms. The evaluation results demonstrate significant achievements: rule semantic coherence reached $\approx 86\%$, FlowBERT classification accuracy remained consistently above 91%, and the adaptive response module achieved a mitigation success rate of 75.13% with a false positive rate of only 8.78%. These outcomes indicate that T4EDR is not merely conceptual but also practical and highly relevant to addressing the need for adaptive, transparent cyber threat detection and response.

The primary contribution of this study lies in T4EDR's ability to merge the interpretability of security rules with the power of deep learning, resulting in a system that is technically robust while remaining auditable under modern security governance. The findings firmly establish T4EDR as a viable approach for adoption in cybersecurity practice, as it simultaneously shortens time-to-detect and enhances time-to-respond. Looking forward, this research opens pathways for further development through cross-domain validation on real world production logs, reinforcement learning integration into the response module, and active learning for managing novelty alerts. With this foundation, T4EDR holds strong potential to evolve into a generative framework that supports

organizational resilience at both national and international levels.

ACKNOWLEDGMENT

The author would like to thank Muhammadiyah University Makassar for its academic support and conducive environment, as well as COCONUT Computer Club for its collaboration and constructive technical discussions. Appreciation is also given to open source communities such as Wazuh and CIC-IDS2018, which provided an important foundation for the experiments and validation of this research.

REFERENCES

- [1] Karantzas, G.; Patsakis, C. (2021). An Empirical Assessment of Endpoint Detection and Response Systems against APTs. *J. Cybersecur. Priv.*, 1(3), 387–421, doi:[10.3390/jcp1030021](https://doi.org/10.3390/jcp1030021).
- [2] Kaur, Harpreet & SL, Dharani & Paul, Tirtharaj & Thakur, Rohit & Reddy, K. & Mahato, Jay & Naveen, Kaviti. (2024). Evolution of Endpoint Detection and Response (EDR) in Cyber Security: A Comprehensive Review. *E3S Web of Conferences*. 556. 01006, doi: [10.1051/e3sconf/202455601006](https://doi.org/10.1051/e3sconf/202455601006).
- [3] Ponemon Institute. (2020). The State of Endpoint Security Risk. Ponemon Research Report. [Online]. <https://ponemonsullivanreport.com/2020/05/>.
- [4] Scarfone, Karen & Mell, Peter. (2007). NIST Special Publication 800-94, Guide to Intrusion Detection and Prevention Systems (IDPS), doi: [10.6028/nist.sp.800-94](https://doi.org/10.6028/nist.sp.800-94).
- [5] Moustafa, N.; Slay, J. (2015). UNSW-NB15: A comprehensive dataset for NIDS. *MilCIS* 2015, doi: [10.1109/MilCIS.2015.7348942](https://doi.org/10.1109/MilCIS.2015.7348942).
- [6] Shone, N.; Ngoc, T.N.; Phai, V.D.; Shi, Q. (2018). A deep learning approach to network intrusion detection. *IEEE TETCI*, 2(1), 41–50, doi: [10.1109/TETCI.2017.2772792](https://doi.org/10.1109/TETCI.2017.2772792).
- [7] Kim, G.; Lee, S.; Kim, S. (2016). A novel hybrid intrusion detection method integrating anomaly detection with misuse detection. *Expert Syst. Appl.*, 41(4), 1690–1700, doi: [10.1016/j.eswa.2013.08.066](https://doi.org/10.1016/j.eswa.2013.08.066).
- [8] M. Thamrin et al., Machine learning for global trade analysis: a hybrid clustering approach using DBSCAN, elbow, and SOM, *IAES Int. J. Artif. Intell.*, vol. 14, no. 4, p. 3033, Aug. 2025, doi:

- [10.11591/ijai.v14.i4.pp3033-3046](https://doi.org/10.11591/ijai.v14.i4.pp3033-3046).
- [9] Wanjau, Stephen & Wambugu, Geoffrey & Oirere, Aaron. (2022). Network Intrusion Detection Systems: A Systematic Literature Review of Hybrid Deep Learning Approaches. *International Journal of Emerging Science and Engineering*. 10. 1-16, doi: [10.35940/ijese.F2530.0610722](https://doi.org/10.35940/ijese.F2530.0610722).
 - [10] Ryan Sheatsley, Nicolas Papernot, Michael J. Weisman, Gunjan Verma, and Patrick McDaniel. 2022. Adversarial examples for network intrusion detection systems. *J. Comput. Secur.* 30, 5 (2022), 727–752. doi: [10.3233/JCS-210094](https://doi.org/10.3233/JCS-210094).
 - [11] Wazuh. (2023). Wazuh Documentation: Security rules and decoders. [Online]. <https://documentation.wazuh.com>.
 - [12] Zouhair Chiba, Noredine Abghour, Khalid Moussaid, Amina El omri, and Mohamed Rida. 2019. Intelligent approach to build a Deep Neural Network based IDS for cloud environment using combination of machine learning algorithms. *Comput. Secur.* 86, C (Sep 2019), 291–317, doi: [10.1016/j.cose.2019.06.013](https://doi.org/10.1016/j.cose.2019.06.013).
 - [13] A. R. Wahid, D. Anggreani, M. A. M. Hayat, A. Abd Rahman, and M. Faisal, “Stacking architecture-endpoint detection: a hybrid multi layered architecture for endpoint threat detection, *Int. J. Adv. Appl. Sci.*, vol. 14, no. 4, p. 1263, Dec. 2025, doi: [10.11591/ijaas.v14.i4.pp1263-1280](https://doi.org/10.11591/ijaas.v14.i4.pp1263-1280).
 - [14] Manocchio, L.D. et al. (2023). FlowTransformer: A Transformer framework for flow-based IDS. *arXiv:2304.14746*.
 - [15] Koukoulis, I.; Syrigos, I.; Korakis, T. (2025). Self-Supervised Transformer-based Contrastive Learning for IDS. *arXiv:2505.08816*.
 - [16] Zhang, C.; Li, J.; Wang, N.; Zhang, D. (2025). Transformer + CNN-BiLSTM IDS for IoT. *Sensors*, 25(9), 2725, doi:[10.3390/s25092725](https://doi.org/10.3390/s25092725).
 - [17] Alshomrani, M. (2024). Survey of Transformer-based Malicious Software Detection System. *Electronics*, 13(23), 4677, doi:[10.3390/electronics13234677](https://doi.org/10.3390/electronics13234677).
 - [18] M. Faisal, T. K. A. Rahman, I. Mulyadi, K. Aryasa, Irmawati, and M. Thamrin, A Novelty Decision-Making Based on Hybrid Indexing, Clustering, and Classification Methodologies: An Application to Map the Relevant Experts Against the Rural Problem, *Decis. Mak. Appl. Manag. Eng.*, vol. 7, no. 2, pp. 132–171, Feb. 2024. doi: [10.31181/dmame7220241023](https://doi.org/10.31181/dmame7220241023).
 - [19] MITRE Corporation. (2023). ATT&CK Framework. [Online]. <https://attack.mitre.org>.
 - [20] Tounsi, Wiem & Rais, Helmi. (2017). A survey on technical threat intelligence in the age of sophisticated cyber attacks. *Computers & Security*. 72, doi: [10.1016/j.cose.2017.09.001](https://doi.org/10.1016/j.cose.2017.09.001).
 - [21] M. Faisal et al., A Hybrid MOO, MCGDM, and Sentiment Analysis Methodologies for Enhancing Regional Expansion Planning: A Case Study Luwu - Indonesia, *Int. J. Math. Eng. Manag. Sci.*, vol. 10, no. 1, pp. 163–188, Feb. 2025, doi: [10.33889/IJMEMS.2025.10.1.010](https://doi.org/10.33889/IJMEMS.2025.10.1.010).
 - [22] Alotaibi, Afran & Rassam, Murad. (2023). Adversarial Machine Learning Attacks against Intrusion Detection Systems: A Survey on Strategies and Defense. *Future Internet*. 15. 62, doi: [10.3390/fi15020062](https://doi.org/10.3390/fi15020062).
 - [23] Linan Huang, Quanyan Zhu. (2020). A dynamic games approach to proactive defense strategies against Advanced Persistent Threats in cyber-physical systems . *Computers & Security* Vol 89, doi: [10.1016/j.cose.2019.101660](https://doi.org/10.1016/j.cose.2019.101660).
 - [24] Fazeldelkordi, Elahe & Grønli, Tor-Morten. (2022). A Survey of Security Architectures for Edge Computing-Based IoT. *IoT*. 3. 332-365, doi: [10.3390/iot3030019](https://doi.org/10.3390/iot3030019).
 - [25] Xie, Yafei. (2023). Machine learning-based DDoS detection for IoT networks. *Applied and Computational Engineering*. 29. 99-107, doi: [10.54254/2755-2721/29/20230972](https://doi.org/10.54254/2755-2721/29/20230972).
 - [26] Jiang, Yuning & Meng, Qiaoran & Shang, Feiyang & Oo, Nay & Minh, Le & Lim, Hoon & Sikdar, Biplab. (2025). MITRE ATT&CK Applications in Cybersecurity and The Way Forward, doi: [10.48550/arXiv.2502.10825](https://doi.org/10.48550/arXiv.2502.10825).
 - [27] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser and Illia Polosukhin. (2017). Attention Is All You Need. *arXiv:1706.03762v7*.